Red Hat OpenShift AI Self-Managed 2.25

Working with the model catalog

Working with the model catalog in Red Hat OpenShift AI Self-Managed

Last Updated: 2025-10-28

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Legal Notice

Abstract

Discover and evaluate generative AI models in the model catalog and select models to register, deploy, and customize.

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PREFACE

As a data scientist in OpenShift AI, you can discover and evaluate the generative AI models that are available in the model catalog. From the model catalog, you can select the models that you want to register, deploy, and customize.

CHAPTER 1. OVERVIEW OF MODEL REGISTRIES AND MODEL CATALOG

A model registry acts as a central repository for administrators and data scientists to register, version, and manage the lifecycle of AI models before configuring them for deployment. A model registry is a key component for AI model governance.

The model catalog provides a curated library where data scientists can discover and evaluate the available generative AI models to find the best fit for their use cases.

1.1. MODEL REGISTRY

A model registry is an important component in the lifecycle of an artificial intelligence/machine learning (AI/ML) model, and is a vital part of any machine learning operations (MLOps) platform or workflow. A model registry acts as a central repository, storing metadata related to machine learning models from development to deployment. This metadata ranges from high-level information like the deployment environment and project, to specific details like training hyperparameters, performance metrics, and deployment events.

A model registry acts as a bridge between model experimentation and serving, offering a secure, collaborative metadata store interface for stakeholders in the ML lifecycle. Model registries provide a structured and organized way to store, share, version, deploy, and track models.

OpenShift AI administrators can create model registries in OpenShift AI and grant model registry access to data scientists. For more information, see Managing model registries.

Data scientists with access to a model registry can use it to store, share, version, deploy, and track models. For more information, see Working with model registries.

1.2. MODEL CATALOG

Data scientists can use the model catalog to discover and evaluate the models that are available and ready for their organization to register, deploy, and customize.

The model catalog provides models from different providers that data scientists can search and discover before they register models in a model registry and deploy them to a model serving runtime. OpenShift AI administrators can configure the available repository sources for models displayed in the model catalog.

OpenShift AI provides a default model catalog, which includes models from providers such as Red Hat, IBM, Meta, Nvidia, Mistral AI, and Google.

For more information about how data scientists can use the model catalog, see Working with the model catalog.

CHAPTER 2. VIEWING MODELS IN THE MODEL CATALOG

You can discover and evaluate the available generative AI models in the model catalog to find the best fit for your use cases.

Prerequisites

- You are logged in to Red Hat OpenShift AI.
- The model registry component is enabled in your OpenShift AI deployment. For more information, see Enabling the model registry component.

Procedure

1. From the OpenShift AI dashboard, click **Models → Model catalog**.
2. The **Model Catalog** page provides a high-level view of available models, including the model name, description, and labels such as task, license, and provider.
3. In the drop-down list, select from the available catalog sources that have been configured by your administrator. The **Default Catalog** is displayed by default.

   NOTE

   OpenShift cluster administrators can configure additional model catalog sources. For more details, see the Kubeflow Model Registry community documentation on configuring catalog sources.

4. Use the search bar to find a model in the catalog. You can enter text to search by model name, description, or provider.
5. Click the name of a model to view the model details page. This page displays the model description and the **Model card** information supplied by the model provider. This includes details such as the model’s intended use and potential limitations, training parameters and datasets, and evaluation results.
6. You can click **Load more models** to scroll and view additional models available in the catalog. Repeat this step until all models are loaded.

Verification

- You can view the information about a selected model on the model details page.

CHAPTER 3. REGISTERING A MODEL FROM THE MODEL CATALOG

As a data scientist, you can register models directly from the model catalog and create the first version of the new model.

Prerequisites

- You are logged in to Red Hat OpenShift AI.
- You have access to an available model registry in your deployment.

Procedure

1. From the OpenShift AI dashboard, click **Models → Model catalog**.
2. In the drop-down list, select from the available catalog sources that have been configured by your administrator. The **Default Catalog** is displayed by default.

   > **NOTE**
   >
   > OpenShift cluster administrators can configure additional model catalog sources. For more details, see the Kubeflow Model Registry community documentation on configuring catalog sources.

3. Use the search bar to find a model in the catalog. You can enter text to search by model name, description, or provider.
4. Click the name of a model to view the model details page.
5. Click **Register model**.
6. From the **Model registry** drop-down list, select the model registry that you want to register the model in.
7. In the **Model details** section, configure details to apply to all versions of the model:
   a. Optional: In the **Model name** field, update the name of the model.
   b. Optional: In the **Model description** field, update the description of the model.
8. In the **Version details** section, enter details to apply to the first version of the model:
   a. In the **Version name** field, enter a name for the model version.
   b. Optional: In the **Version description** field, enter a description for the first version of the model.
   c. In the **Source model format** field, enter the name of the model format, for example, **ONNX**.
   d. In the **Source model format version** field, enter the version of the model format.
9. In the **Model location** section, the URI of the model is displayed.
10. Click **Register model**.

Verification

- The new model details and version are displayed on the **Overview** tab on the model details page.
- The new model and version are displayed on the **Model registry** page.

CHAPTER 4. DEPLOYING A MODEL FROM THE MODEL CATALOG

You can deploy models directly from the model catalog.

> **NOTE**
>
> OpenShift AI model serving deployments use the global cluster pull secret to pull models in ModelCar format from the catalog.
>
> For more information about using pull secrets in OpenShift, see Updating the global cluster pull secret in the OpenShift documentation.

Prerequisites

- You have completed the prerequisites in Deploying models on the single-model serving platform.
- The model registry component is enabled in your OpenShift AI deployment. For more information, see Enabling the model registry component.

Procedure

1. From the OpenShift AI dashboard, click **Models → Model catalog**.
2. In the drop-down list, select from the available catalog sources that have been configured by your administrator. The **Default Catalog** is displayed by default.

   > **NOTE**
   >
   > OpenShift cluster administrators can configure additional model catalog sources. For more details, see the Kubeflow Model Registry community documentation on configuring catalog sources.

3. Use the search bar to find a model in the catalog. You can enter text to search by model name, description, or provider.
4. Click the name of a model to view the model details page.
5. Click **Deploy model** to display the **Deploy model** dialog.
6. From the **Project** drop-down list, select a project in which to deploy your model.

   > **NOTE**
   >
   > Models using OCI storage can only be deployed on the single-model serving platform. Projects using the multi-model serving platform do not appear in the project list.

7. In the **Model deployment** section:
   a. Optional: In the **Model deployment name** field, enter a unique name for your model deployment. This field is autofilled with a value that contains the model name by default.

This is the name of the inference service created when the model is deployed.

b. Optional: Click **Edit resource name**, and then enter a specific resource name for the model deployment in the **Resource name** field. By default, the resource name matches the name of the model deployment.

> **IMPORTANT**
>
> Resource names are what your resources are labeled as in OpenShift. Your resource name cannot exceed 253 characters, must consist of lowercase alphanumeric characters or -, and must start and end with an alphanumeric character. Resource names are not editable after creation.
>
> The resource name must not match the name of any other model deployment resource in your OpenShift cluster.

c. From the **Serving runtime** list, select a model-serving runtime that is installed and enabled in your OpenShift AI deployment. If project-scoped runtimes exist, the **Serving runtime** list includes subheadings to distinguish between global runtimes and project-scoped runtimes.

d. From the **Model framework** list, select a framework for your model.

> **NOTE**
>
> The **Model framework** list shows only the frameworks that are supported by the model-serving runtime that you specified when you deployed your model.

8. From the **Deployment mode** list, select **KServe RawDeployment** or **Knative Serverless**. For more information about deployment modes, see About KServe deployment modes.

a. In the **Number of model server replicas to deploy** field, specify a value.

b. From the **Model server size** list, select a value.

c. If you have created a hardware profile, select a hardware profile from the **Hardware profile** list. If project-scoped hardware profiles exist, the **Hardware profile** list includes subheadings to distinguish between global hardware profiles and project-scoped hardware profiles.

> **IMPORTANT**
>
> By default, hardware profiles are hidden from appearing in the dashboard navigation menu and user interface. In addition, user interface components associated with the deprecated accelerator profiles functionality are still displayed. To show the **Settings → Hardware profiles** option in the dashboard navigation menu and the user interface components associated with hardware profiles, set the **disableHardwareProfiles** value to **false** in the **OdhDashboardConfig** custom resource (CR) in OpenShift. For more information about setting dashboard configuration options, see Customizing the dashboard.

d. In the **Model route** section, select the **Make deployed models available through an external route** checkbox to make your deployed models available to external clients.

e. In the **Token authentication** section, select the **Require token authentication** checkbox to require token authentication for your model server. To finish configuring token authentication, perform the following actions:

   i. In the **Service account name** field, enter a service account name for which the token will be generated. The generated token is created and displayed in the **Token secret** field when the model server is configured.

   ii. To add an additional service account, click **Add a service account** and enter another service account name.

9. In the **Source model location** section, select **Current URI** to deploy the selected model from the catalog.

10. Optional: Customize the runtime parameters in the **Configuration parameters** section:

    a. Modify the values in **Additional serving runtime arguments** to define how the deployed model behaves.

    b. Modify the values in **Additional environment variables** to define variables in the model's environment.

11. Click **Deploy**.

Verification

- The model deployment is displayed on the **Models → Model Deployments** page.
- The model deployment is displayed in the **Latest deployments** section of the model details page.
- The model deployment is displayed on the **Deployments** tab for the model version.

Additional resources

- For more information about deployment options on the single-model serving platform, see Deploying models on the single-model serving platform.